Finance Self-Study Roadmap

Personal finance · Investing · Accounting · AI/Quant — Specific courses, verified instructors

Phase 1 — Foundations — How money actually works

Now → July 2025 · 1–2 hrs/week · Alongside Java prep

Personal Finance & Budgeting

Goal-setting, budgeting (50/30/20), saving, risk, borrowing basics

Coursera (paid) — Single course

Financial Planning for Young Adults Univ. of Illinois · Inst: Nicholas Paulson, Charles Chaffin · 4 weeks

How Credit Works

Credit scores, good vs bad debt, interest, credit card mechanics

Alison (free) — Single course

An Introduction to Personal Finance and Credit alison.com/course/an-introductory-course-in-personal-finance-and-credit

Banking & Compound Interest

How banks work, compound interest, savings accounts, inflation basics

YouTube (free) — Khan Academy

Khan Academy: Finance & Capital Markets khanacademy.org → Personal Finance → Banking

Taxes 101

W-2 vs 1099, brackets, deductions, Form 1040 hands-on, withholding

Coursera (paid) — Single course

Federal Taxation I: Individuals, Employees & Sole Proprietors Univ. of Illinois · 4.8 stars · Updated for 2025 tax law

Phase 2 — Investing — Making money work for you

July – December 2025 · After starting McDaniel · 2 hrs/week

Stock Market Fundamentals

Risk, CAPM, behavioral finance, insurance, banking, investment banking

Coursera (paid) — Single course, auditable free

Financial Markets Yale University · Prof. Robert Shiller (Nobel laureate) · 4.8 stars · 33 hrs

Investing Essentials: Stocks, ETFs & Funds

Stocks, bonds, ETFs, REITs, passive vs active, portfolio building

Udemy (library card) — Single course

Complete Investing Course 2026 (Stocks, ETFs Investing) Inst: Calvin K. Lee, CPA/CA/MBA · York Univ. · 74,000+ students

Portfolio Theory & Management

Asset allocation, risk/return, diversification, rebalancing, backtesting

Coursera (paid) — 4-course Specialization

Investment and Portfolio Management Specialization Rice University · Inst: Arzu Ozoguz + faculty · 15 weeks + capstone

Real Estate Basics

Mortgages, home equity, property ownership, financing, REITs

Alison (free) — Single course

Introduction to Real Estate Investing alison.com/course/introduction-to-real-estate-investing

Phase 3 — Accounting & Business Finance

Early 2026 · Retroactively explains what Morgan Stanley systems were actually doing

Accounting Fundamentals

Income statement, balance sheet, cash flow; GAAP; reading real 3M financials

Coursera (paid) — Single course (part of Wharton Business Foundations)

Introduction to Financial Accounting UPenn / Wharton · Prof. Brian J. Bushee · 4.8 stars

Reading Financial Reports

10-K analysis, ratio analysis, red flags, SEC filings, earnings quality

Udemy (library card) — Single course

Practical Financial Statement Analysis Inst: Jason Williams (ex-Credit Suisse Accounting & Tax Research)

Corporate Finance

Risk/return, CAPM, cost of capital, DCF, NPV, project evaluation, value creation

Coursera (paid) — Single course

Corporate Finance Essentials IESE Business School · Prof. Javier Estrada (PhD Illinois) · 4.8 stars · 13 hrs

FinTech Context

Blockchain, crypto, payments, robo-advising, P2P lending, AI in finance

Coursera (paid) — 4-course Specialization

FinTech: Foundations & Applications of Financial Technology UPenn Wharton · Profs. Geczy, Sarin, Wachter, Musto · 60,500 enrolled

Phase 4 Advanced — Finance meets your AI/ML track

2026+ · Synergizes directly with McDaniel AI coursework and Python skills

ML for Quantitative Trading

Trading strategies, pairs/momentum trading, Keras/TensorFlow models, backtesting

Coursera (paid) — 3-course Specialization

Machine Learning for Trading Specialization Google Cloud + NYIF · Inst: Jack Farmer · Intermediate · 4 weeks

Behavioral Economics & Finance

Cognitive biases, irrational decision-making, market anomalies

Coursera (paid) — Single course

Behavioral Finance Duke University · Inst: Emma Rasiel & John A. Forlines III

Debt Management Deep Dive

Debt repayment strategies, long-term planning, financial safety nets

Alison (free) — Single course

Debt Management and Financial Planning alison.com/course/debt-management-and-financial-planning

Python for Finance & Algo Trading

NumPy, Pandas, Matplotlib, QuantConnect/LEAN engine, algorithmic strategies

Udemy (library card) — Single course

Python for Finance and Algorithmic Trading with QuantConnect Inst: Jose Portilla (Head of Data Science, Pierian Training)

Platforms: Coursera (paid subscription) · Udemy (NJ library card) · Alison (free) · Khan Academy / YouTube (free)